Model Risk Factory

Challenge 04 · EXL × UCD Data & AI Hackathon 2026

Agentic model-risk factory: document a legacy model estate, migrate one model from R to Python, validate it, and record who reviewed each decision.

- Three legacy R models documented once, from code and stale notes, with sources, hashes and the Unknowns that stay Unknown.
- One model (MDL_CREDIT_V1) migrated to Python by a real gateway Agent, proven equivalent record by record.
- Performance, calibration, stability and fairness computed by Python. The Agent selects a review focus and Gate C binds the rendered report.
- Each metric value resolves to a JSON path. Review decisions bind the exact artifact hash.

this deck states results only from runs recorded in the repository. Synthetic drill = dev_runs/v2-20261110T182506Z-9acb1487; UCI history = v2-20261109T210903Z-3dcfab73 (seed 46, historical). No formal v3 run exists: release is BLOCKED (slide 9).

The client problem

CLIENT-CONTEXT (from the challenge brief, not measured here): a bank runs ~250 models; documentation drifts away from code; a model review takes 8–12 weeks; PRA SS1/23 sets the expectation that models are documented, validated, monitored and independently reviewed.

What that costs in practice, and what this prototype addresses:

- The document and the code disagree, and nobody can say which one is right without reading both.
- Re-implementation is checked by eye, sampled, or "the AUC looks fine".
- Validation evidence is rebuilt per review instead of being produced by the pipeline.
- Drift is found by a spreadsheet refresh, not by a metric with a recorded threshold.

docs/01_比赛背景与官方要求.md and the official PDF. The 250-model / 8–12-week figures are client backdrop only and are never presented as this project's results.

Method and architecture

The Agent proposes artifacts, deterministic checks validate them, and reviewers record decisions at three gates.

- Discover → split the UCI extract once, lock seed, hashes and tolerances.
- Document → Agent reads R code + simulated note, returns a schema-validated inventory; Gate A approves the artifact hash.
- Redevelop → Agent writes the Python candidate; Gate B approves the exact 64-hex source hash before anything runs.
- Validate → candidate executes in a locked-down container; a trusted validator compares it with the R baseline.
- Monitor → PSI-based drift state, calibration table, fairness audit; Gate C approves the rendered report.
- Hard boundary: Holdout inputs and answers never enter the Agent context; only aggregate, desensitised metrics do.

architecture detail in materials/architecture.md. Gate definitions: docs/02_项目SOP.md §7.4.

Inventory and documented gaps

Three models documented from their real code and their real (deliberately stale) notes.

- MDL_CREDIT_V1: code threshold 0.35 vs documented 0.50 — kept as a conflict, never auto-normalised; 14 verified sources.
- MDL_CREDIT_V2 and MDL_CREDIT_TREE: undocumented input limits, transformations and hyperparameters recorded as conflicts; owner and training date stay Unknown.
- Each confirmed conflict carries the file path, line, excerpt and SHA-256 from the R code and legacy note.
- The allowed findings are model-specific. Equal code and document thresholds cannot be recorded as a threshold conflict.
- HIST-UCI: 3 inventories, 3 Gate A approvals, 4 Unknown fields each.

HIST-UCI model_inventory.json, inventories/.json, inventory_validation.json; rule set src/inventory.py + config/inventory_expectations.json; regression tests/test_v3_gap_inventory_conflicts.py.*

R to Python parity

The candidate reuses the exported R coefficients, medians, feature order and the 0.35 threshold. Nothing is retrained.

- HIST-UCI: 6,000 development records and 6,000 Holdout records compared one by one — max |Δp| 2.220e-16, mean |Δp| 2.707e-17, 0 classification mismatches, 0 behaviour mismatches (tolerance 1e-8 / 1e-10).
- Behaviour is tested too: missing values, imputation flags, invalid rows and rejection codes match record by record, plus row-order, column-order and structural-refusal fixtures.
- A deliberately altered coefficient (+0.001 on delay_1) is caught by the fixed validator; the original baseline and test hashes stay unchanged.
- The Agent never sees the Holdout: it only gets development parity summaries and aggregate diagnostics.

HIST-UCI parity//parity_summary.json, mutation/mutation_summary.json, dev/attempt_000/fixtures/; thresholds config/model_config.json.*

Validation and monitoring

Python computes the metrics and renders the conclusions. The Agent selects a review focus and its structured facts are checked against the metrics.

- Performance (HIST-UCI Holdout, 6,000 rows): ROC-AUC 0.7529, Brier 0.1422, decile calibration table, confusion matrix at p ≥ 0.35 (TP 621 / FP 501 / FN 701 / TN 4177).
- Stability: weighted drift batch vs dev reference — prediction PSI 0.0170, worst feature PSI 0.0799 (util_1); attention 0.10 / review 0.25 are project demo rules, not a regulatory standard.
- Fairness audit: age_band 21–25 TPR 0.5477 vs reference 0.4392 (+0.1086), sex=1 TPR difference −0.0387; small groups are marked LOW_SAMPLE and rates are not assessable.
- Limits are printed next to the numbers: same-period hold-out (not a back-test), weighted drift is a sensitivity analysis, cross-time back-test unavailable.
- SYN-SEED47: 196 report values, 0 unresolved, Gate C re-renders and compares every cell.

HIST-UCI validation_metrics.json, report/validation_report.md; SYN-SEED47 report/values_index.csv, report/review_manifest.json.

Measured automation time

These timings cover automation on the synthetic workflow.

- SYN-SEED47 full pipeline (3 inventories + migration + validation + report + export): 29 CLI commands, 200.5 s of command time, 3.49 min wall clock, 0 failed steps.
- Agent stages inside that: inventory 12.7 s / 31.6 s / 40.3 s per model, candidate 66.1 s, report summary 9.8 s (gateway deepseek-v4.1-flash:1017).
- Historical approvals in this drill were scripted. Manual and Agent-plus-human paired measurements are pending.
- materials/impact_measurements.csv holds the paired measurement protocol and the measured Agent rows; manual rows are PENDING.
- Batch path to ~250 models: same gates, same validator, per-model context and one Holdout per campaign — capacity is a queue question, not a redesign.

SYN-SEED47 command_log.jsonl timings, agent_audit_log.jsonl, provider/request_log.jsonl; materials/impact_measurements.csv + its README.

Responsible AI

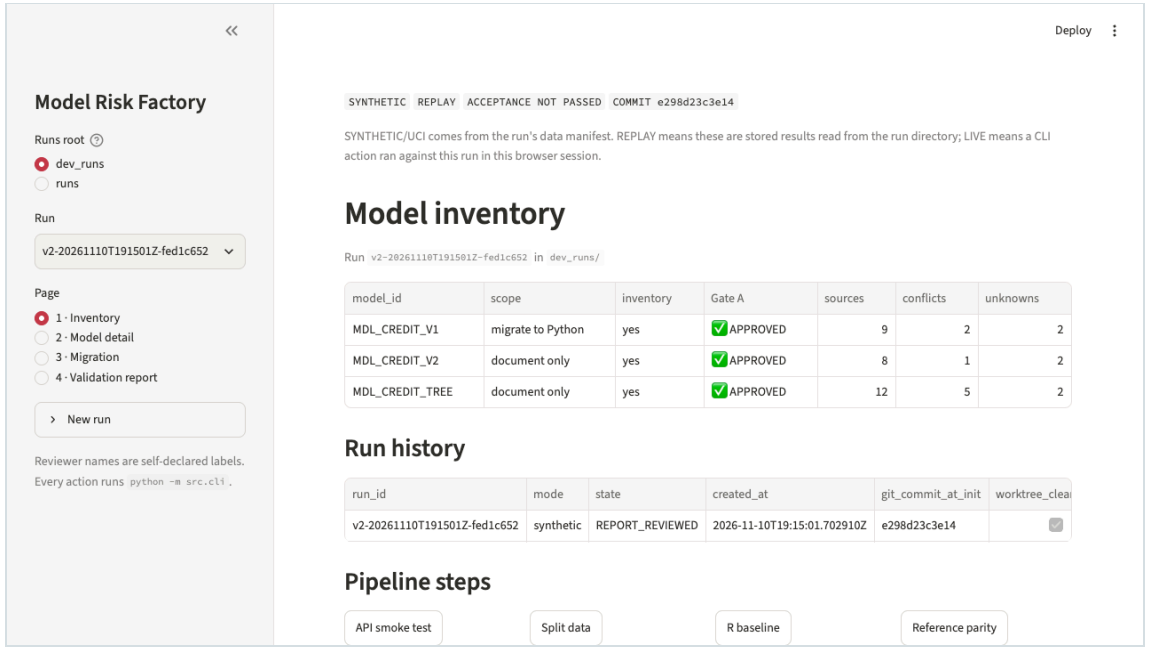
- Data: UCI Default of Credit Card Clients (CC BY 4.0), a 2005 Taiwan sample; no direct identifiers but demographic attributes exist. Competition use permission is still recorded as unconfirmed (uci_competition_use_approved=false).
- Privacy: demographics are audit-only and never model inputs; no raw rows, labels or Holdout content are sent to the LLM — only R code, the simulated note, schemas and aggregate diagnostics; API keys live in local environment variables only.
- Bias: TPR/FPR by age_band and sex are audit signals, not a fairness verdict; small groups are marked not assessable; the model is not claimed to be fair.
- Limitations: not production-ready, not independently validated, not a regulatory certification; no cross-time back-test; report conclusions come from computed metrics and are checked again at Gate C.
- Safeguards: generated code runs only in a no-network, read-only, non-root container with memory/CPU/PID/time limits; gates A/B/C bind to hashes and invalidate on any source, config or candidate change.

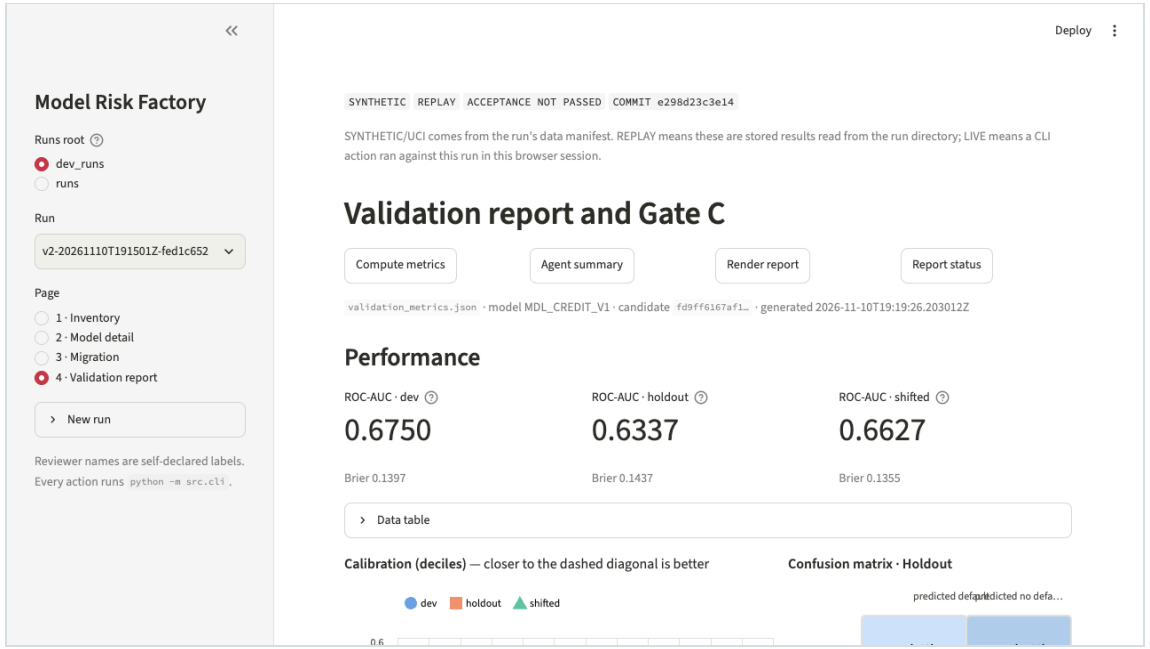
HIST-UCI boundary/boundary_report.json, review_log.json, report/validation_report.md; materials/responsible_ai_note.md.

Prototype walkthrough and release state

Four Streamlit pages over the same CLI the pipeline uses — every button runs python -m src.cli, so the UI cannot bypass a gate.

- Inventory, Model detail (Gate A), Migration (Gate B + Docker), Validation report (Gate C + downloads).
- The 1:51 demo replays a synthetic run. The current drill pauses at every gate for a reviewer decision.
- Release state, stated plainly: v1 released (v1 → 70858cd, 7/7 PASS). v2 BLOCKED — its evidence was produced on a dirty worktree and matches no commit; release-check returns 17 gaps. v3 has no formal run: seed 47 needs a confirmed UCI permission, ledger confirmation and an authorised approver.





docs/v3/00_baseline_status.md, docs/v3/03_release_readiness.md, docs/v3/02_new_environment.md; screenshots in docs/v3/ui_screens/.

Roadmap and disclosure

Next for the estate, and what was actually used to build this.

- Short term: one authorised formal seed-47 run on the frozen commit, human Gate A/B/C sign-off, recorded demo of the same chain.
- Then: SAS/other formats behind the same inventory contract; batch queue across ~250 models; MLflow/Azure ML as an evidence sink, not a replacement for the gates; genuine out-of-time data for back-testing.
- AI tools used: Claude Code / Kimi Code CLI (implementation, tests, docs) and this delivery Agent; in-product gateway sa.kaiwen.dev with deepseek-v4.1-flash:1017 for inventory extraction, candidate generation and report explanation (v1 candidate used kimi-k2.7-code).
- Existing code and data: R glm/rpart/jsonlite, Python NumPy/pandas/Pydantic/pytest/Docker/Streamlit, the UCI dataset, and the three team-built R models plus their simulated notes.
- What is human-written: the trusted Python reference predictor and the fixed tests. What is Agent-written: the migration candidate, the inventories and the report narrative.

README.md (AI tool and existing-code disclosure), HIST-UCI provider/request_log.jsonl, agent_audit_log.jsonl, V1-UCI candidate.json.